Sean Tran

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Education

The University of Texas at Dallas	**B.S. in Physics** (Expected Dec 2026)	*2024 – Present*
	B.S. in Computer Science	*2019 – 2024*

Relevant Experience

NSF REU Undergraduate Researcher — Computational Astrophysics *May 2025 – June 2026*
University of Texas at Dallas – Richardson, TX

- Developed and optimized Python-based simulations for gravitational lensing, executing large-scale parameter sweeps across high-dimensional spaces to improve computational efficiency
- Applied chi-squared minimization and Bayesian inference to evaluate hundreds of thousands of model configurations, calibrating parameters and identifying sensitivities to input variations
- Implemented numerical ray-tracing algorithms for modeling optical systems under varying condition, extending models toward wave-based effects and using image-based visualization to interpret system behavior
- Presented data-driven results at UTD SPUR 2025; selected for research award
- Tools Used: *Python (NumPy, SciPy, Astropy, Matplotlib, Plotly), MATLAB, Linux, Bash, Jupyter, Git, Glafic*

Restaurant Manager — Family-Owned Business *June 2016 – Present*
Golden Deli – Addison, TX

- Managed operations including transactions, inventory control, and equipment maintenance while troubleshooting issues to maintain efficiency and ensure exceptional customer service

Projects and Builds

AES Encryption Engine & Visualization Tool

- Restructured the Advanced Encryption Standard symmetric key algorithm recursively in Java to emulate OpenSSL in a more user-friendly environment, resulted in stronger foundation for lectures and future topics
- Employs multi-use functions to and block-cipher chaining to increase runtime efficiency while displaying step-by-step breakdown to provide user with better understanding of encryption process

Semiconductor Characterization — Hall Effect Measurements

- Determined carrier density, mobility, and conductivity of doped semiconductors by combining Hall effect measurements with transport analysis, validating results against expected material behavior
- Calculated semiconductor band gap ($\sim$0.75 eV) from temperature-dependent conductivity using $\ln(\sigma)$ vs $1/T$ analysis, achieving agreement within 12% of accepted values

Laser Diode Spectroscopy — Hyperfine Transitions of Rubidium

- Diagnosed and corrected a systematic measurement error in a laser setup caused by unintended optical paths in the Michelson interferometer, reducing error from an order-of-magnitude deviation to $< 5\%$
- Developed a fringe-based frequency calibration to extract GHz-scale spectral features and determine hyperfine peak separations consistent with known atomic transitions

Optical Schlieren Imaging System — Density Gradients in Gas Propulsion

- Designed and built an optical Schlieren imaging system to capture refractive index variations in gas flow, enabling visualization of otherwise undetectable flow structures
- Aligned and optimized optical components (mirrors, a collimated light source, and knife-edge cut-off) to optimize system sensitivity, spatial resolution, and image contrast

Skills

Programming: Java, Python, C++/C/C#, Bash, SQL, HTML/CSS,
Software: Kali Linux (WSL), Ubuntu, MATLAB, Jupyter, MS Excel, FreeCAD, Arduino IDE, AWS, Firebase, MySQL
Fabrication: Repair and Maintenance, Hand and Power Tools, Basic Electrical Wiring and Circuit Assembly, Soldering